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Microsporogenic analysis on some varieties of *Stevia rebaudiana* Bertoni

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Abstract : In the present investigation microsporogenic studies were done in three varieties of *Stevia rebaudiana* . Meiotic analysis were done using young flower buds. Formation of eleven bivalents were recorded, which were regular in ring and rod shape. There was a negative correlation between bivalents in relation to inter chromosomal distribution of chiasmata.

Key Words: ***Stevia rebaudiana,*** **Meiotic, Bivalents, Chiasmata.**

INTRODUCTION

Stevia botanically known as *Stevia rebaudiana* belongs to family Asteraceae. *Stevia*, a natural sweetening stevioside extract from the *Stevia rebaudiana* plant is also known by the names of sweet leaf or sugar leaf. It is a plant that produces a variety of high potency low calorie sweeteners in its leaf tissue.There are number of therapeutic applications of *Stevia.*

MATERIALS AND METHODS

Var. SRB-123, Var. SRB-128 and Var. SRB-512 of *Stevia rebaudiana* Bertoni were selected for meiotic studies collected from IKGVKV, Raipur, BAU, Kanke and KGVK, Rukka . Young floral buds collected from their respective plants and were immediately fixed in Carnoy fluid 1:3 acteo-alcohol solution.[1], being transferred to 70% alcohol and stored in refrigeration. The slides were prepared using the technique of squashing the anther, using. 2% aceto-carmine. Microphotographs of PMC's were taken. The data were recorded from 100 pollen mother cells at diakinetic metaphase I stage. The mean square of chiasmata between and within nucleus and the analysis of variance was calculated on the principles earlier proposed[2].

More than 500 pollen grains per varieties were stained with 2% aceto-carmine. The pollen grains were considered fertile which took up stain and those which did not stained were sterile. Percentage of pollen fertility and sterility were calculated and diameter of pollen grains were measured.

RESULT AND DISCUSSION

The results obtained by meiotic analysis were presented in table 1 and figure 1to 3. Normal meiotic division occurred in all the three varieties and regular formation of eleven bivalents in ring and rod shape. The number of bivalents were constant and comparatively smaller in size. Relatively smaller meiotic chromosomes in Asteraceae were earlier reported[3]. All the varieties showed the gametic number (n) and basic gametic value as eleven (x = 11) were earlier also reported[4] were confirmed. Univalents, trivalents, tetravalents and multivalents were absent. PMC showed groups with different peculiarities in cells at Metaphase I. The percentage of ring bivalents with two chiasmata were considerably high which might be the out come of maximum homologous chromosomes. The variation in nature of ring and rod bivalents might be due to the structural differences in their karyotype or due to changes

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in the nature of gene or set of genes controlling the chiasma formation. The meiotic division was non – synchronized i.e. all the stages of meiotic division were present in a single anther but micro and macro lesion which occurs spontaneously in the chromosomes were altogether absent which were also confirmed[5]. This indicates that homologous chromosome fidelity was synchronous with that of synapsis giving size to ring bivalents. A negative correlation co – efficient existed in all the three varieties. This indicates that the mean square values between nuclei in comparison to that of within nuclei were higher and therefore, variations existed in the number of chiasmata per bivalent even in the same nucleus. The values of mean square within nuclei indicates that competition also existed among the bivalents of the same pollen mother cells in the formation of certain number of chismata until it reaches a saturation point. It also agrees with the views in Hypothesis of competition[2]. The difference in variance ratio of the varieties investigated clearly suggested that there are variation in size, nature and over all karyotypic make up of the bivalents in the same pollen mother cell.

The result for pollen studies were depicted in Table–2 and Fig.–4. The maximum fertility percentage was recorded 62.83 in var. SRB – 512. Where as it was minimum (37.16%) in the same variety i.e. var. SRB – 512. There were much variation in pollen viability in family Asteraceae (Compositae). The pollen diameter was recorded 21.54 µm, 21.69 µm and 21.72 µm in var. SRB – 128, var. SRB – 123 and SRB – 512 respectively. The diameter of the pollen grains were almost equal in these three varieties.The normal meiosis and high rates of tetrad normality inferred from the pollen viability. The pairing at diakinesis was normal. It suggested that in normal conditions the viability of pollen were not affected by external factor. Most member of Asteraceae showed low pollen viability[6,7] and considerable variation in pollen size[8], while few showed high pollen viability[9]. Pollen fertility leads to increase in yield[10].The low pollen viability suggests that plant produce inviable seeds, so *Stevia* plants were mostly propagated vegetatively.

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Table-1 : Meiotic analysis and Chiasmata frequency in different varieties of *Stevia rebaudiana* Bertoni

	Varieties	No. of Cells Observed	No. of Rings	No. of Rods	Mean X^{ta}	½ X^{ta}	Mean square between nuclei	Mean square within nuclei	Variance ratio	Correlation co-efficient
Meiotic analysis	SRB – 123	100	1596	302	18.98	0.86	13.088	3411.08	0.007	-0.01
	SRB – 128	100	1586	307	18.93	0.86	12.047	3393.32	0.007	-0.01
	SRB – 512	100	1592	304	18.96	0.86	13.804	3403.98	0.008	-0.01

	Varieties	No. of Cells Observed	No. of bivalent with Chiasmata in cell studied		Total Chiasmata	Average Chiasmata per nucleus	Chiasmata per bivalent	Terminalization co-efficient
			No. of Rings	No. of Rods				
Chiasmata frequency	SRB – 123	100	1596	302	1898	18.98	1.725	84.088
	SRB – 128	100	1586	307	1893	18.93	1.720	83.782
	SRB – 512	100	1592	304	1896	18.96	1.723	83.966

Table-2 : Percentage Pollen Fertility in different varieties of *Stevia rebaudiana* Bertoni

Varieties	Total no. of Pollen grains observed	Fertility %	Sterility %	Pollen grain diameter (µm)
SRB – 123	514	61.86	38.13	21.69 ± 0.19
SRB – 128	511	62.81	37.18	21.54 ± 0.17
SRB – 512	522	62.83	37.16	21.72 ± 0.23

Fig.:- 1-3 PMC showing eleven bivalents at Metaphase I

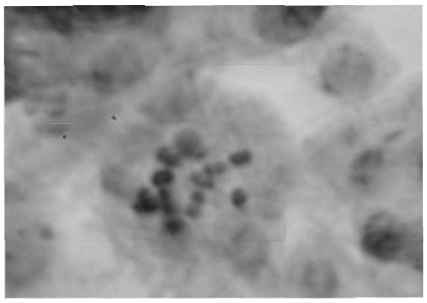


Fig-1:- Var. SRB -123

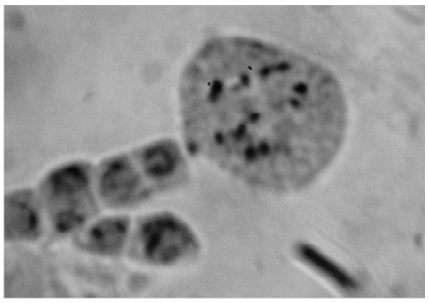


Fig-2:- Var. SRB -128

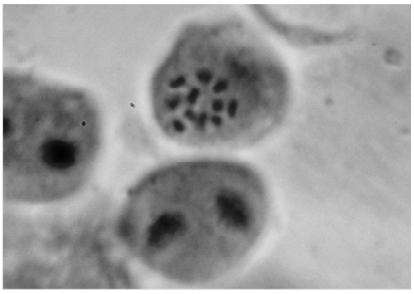


Fig-3:- Var. SRB -512

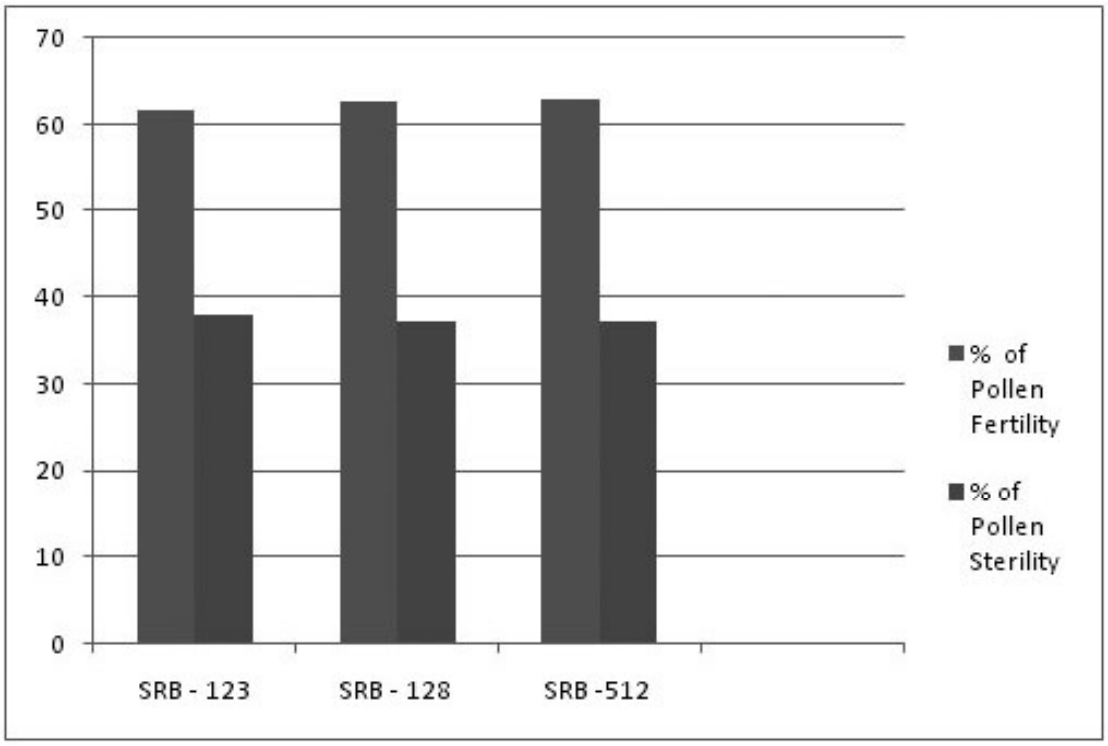


Fig. 4 :- Column graph showing percentage fertility and percentage sterility in different varieties of *Stevia rebaudiana* Bertoni.

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